Associations of Antibiotics, Hormonal Therapies, Oral Contraceptives, and Long-Term NSAIDS With Inflammatory Bowel Disease: Results From the Prospective Urban Rural Epidemiology (PURE) Study

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BACKGROUND & AIMS: **Several medications have been suspected to contribute to the etiology of inflammatory bowel disease (IBD). This study assessed the association between medication use and the risk of developing IBD using the Prospective Urban Rural Epidemiology cohort.**

METHODS: **This was a prospective cohort study of 133,137 individuals between the ages of 20 and 80 from 24 countries. Country-specific validated questionnaires documented baseline and follow-up medication use. Participants were followed up prospectively at least every 3 years. The main outcome was the development of IBD, including Crohn's disease (CD) and ulcerative colitis (UC).**

Abbreviations used in this paper: aOR, adjusted odds ratio; CD, Crohn's disease; IBD, inflammatory bowel disease; NSAID, nonsteroidal anti-inflammatory drug; OR, odds ratio; PURE, Prospective Urban Rural Epidemiology; UC, ulcerative colitis.

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Short-term (baseline but not follow-up use) and long-term use (baseline and subsequent follow-up use) were evaluated. Results are presented as adjusted odds ratios (aORs) with 95% CIs.

RESULTS: **During a median follow-up period of 11.0 years (interquartile range, 9.2–12.2 y), there were 571 incident IBD cases (143 CD and 428 UC). Incident IBD was associated significantly with baseline antibiotic (aOR, 2.81; 95% CI, 1.67–4.73; $P = .0001$) and hormonal medication use (aOR, 4.43; 95% CI, 1.78–11.01; $P = .001$). Among females, previous or current oral contraceptive use also was associated with IBD development (aOR, 2.17; 95% CI, 1.70–2.77; $P < .001$). Nonsteroidal anti-inflammatory drug users also were observed to have increased odds of IBD (aOR, 1.80; 95% CI, 1.23–2.64; $P = .002$), which was driven by long-term use (aOR, 5.58; 95% CI, 2.26–13.80; $P < .001$). All significant results were consistent in direction for CD and UC with low heterogeneity.**

CONCLUSIONS: **Antibiotics, hormonal medications, oral contraceptives, and long-term nonsteroidal anti-inflammatory drug use were associated with increased odds of incident IBD after adjustment for covariates.**

Keywords: Crohn's Disease; NSAIDS; Ulcerative Colitis; Oral Contraceptive; Estrogen; Hormone; IBD; Antibiotics.

The pathogenesis of inflammatory bowel disease (IBD), which includes Crohn's disease (CD) and ulcerative colitis (UC), is believed to involve activation of the intestinal immune system in response to alterations in the gut microbiome among genetically susceptible hosts.[1,2] Environmental factors, such as medications, smoking, stress, air pollution, and diet, have been explored as risk factors for the development of IBD.[3,4] Plausibly, many of these potential risk factors have been shown to cause dysbiosis in the gut microbiome that could predispose to subsequent development of IBD, although the precise cause of IBD is unknown.[5,6]

IBD historically has been regarded as a disease of developed nations, although in the past 2 decades there has been a shift in the epidemiologic pattern of disease. High-income nations with known high prevalence of disease are seeing a stabilization of incident cases, while a rapid increase of incident IBD has been observed in developing nations.[7,8] This suggests that environmental exposures may play a role in mediating the risk of developing IBD. The potential environmental determinants of IBD across various regions is vast, although medications increasingly have been recognized as one broad category of risk factors.

Although nonsteroidal anti-inflammatory drugs (NSAIDs), antibiotics, and oral contraceptives have been well studied as risk factors for IBD, when the totality of evidence is assessed critically, these associations remain largely unclear because of the conflicting body of literature, heterogeneity between studies, and study design limitations.[9,10] Other medications also have been associated with an increased or reduced risk of IBD.[11,12] Given the evolving global epidemiology of IBD, further research across more heterogenous populations with robust design is needed to better understand medications as a risk factor for the development of IBD to validate previously demonstrated relationships, as well as to explore previously unreported associations. This report describes the association between medication use and the risk of developing IBD using the Prospective Urban Rural Epidemiology (PURE) cohort.[4,13,14]

Methods

Study Design

Details regarding the study design and methods of the PURE study have been published previously and are available in the Supplementary Methods section.[13,14]

Outcomes

The primary objective of this study was to evaluate the association between medication use at baseline and the development of IBD (CD or UC) during follow-up evaluation. For baseline medications that were associated significantly with the primary outcome, secondary analyses were planned to evaluate short-term (defined as use only at baseline but not at follow-up evaluation) and long-term use (defined as use at baseline and at next follow-up visit). In addition, tests for heterogeneity were conducted for the outcomes of CD and UC individually. Diagnosis was self-reported by participants.

Statistical Analysis

Descriptive statistics for each comparison are provided. Continuous variables are presented as means (and SDs) or medians (and interquartile ranges). Categoric variables are presented as proportions or percentages and compared using the chi-square test.

Multivariate logistic regression models were used to evaluate the relationship between medications and the outcome of interest. Odds ratios (ORs) and corresponding 95% CIs for the development of the outcome of interest were calculated and minimally adjusted for age,

sex, and center. Fully adjusted comparisons additionally were performed, adjusting for covariates that had a previously identified association with the outcome of interest in the PURE cohort (diet, smoking, body mass index, education, and location) and comorbidities (diabetes, hypertension, stroke, angina/heart attack/coronary artery disease, heart failure, other heart disease, hepatitis/jaundice, cancer, chronic obstructive pulmonary disease, asthma, and tuberculosis).[4] Diet adjustment was performed using the Alternative Healthy Eating Index score.[15] Briefly, the Alternative Healthy Eating Index is an evidence-based tool that evaluates dietary quality and has demonstrated the ability to predict chronic disease risk. Further details of the analyses are provided in the Supplementary Methods section.

Results

A total of 133,137 participants had at least 1 cycle of follow-up evaluation and were included in this study, with the majority having 2 or 3 follow-up evaluations (86,173; 64.7%). During a median follow-up period of 11.0 years (interquartile range, 9.2–12.2 y), a total of 571 participants had a recorded diagnosis of IBD during follow-up evaluation, which included 143 participants with CD and 428 participants with UC. Table 1 summarizes the baseline characteristics of the study population at the time of enrollment. The mean age was 50.3 years (SD, 9.7 y), and nearly 60% of the population was female. A total of 52% of participants resided in urban areas and 58% had at least secondary or higher education, trade, or college/university education. Nearly 18% of the population were current smokers and the mean body mass index was 25.9 (SD, 5.3). Most participants (61.5%) were from low- or lower-middle–income countries. Of those who developed IBD, a total of 310 (54.3%) were from high- or upper-middle–income countries. Of the comorbid conditions evaluated, hypertension was the most prevalent, affecting 11,895 (8.9%) participants, followed by diabetes (3969; 3%), and angina/heart attack/coronary artery disease (2888; 2.2%). Overall, baseline characteristics were similar among the 571 participants who developed IBD compared with those who did not.

Baseline Medication Use and the Development of Inflammatory Bowel Disease

A list of medications and their categories are provided in Supplementary Table 1. Table 2 and Supplementary Tables 2 and 3 show the proportions of participants who reported regular medication use at baseline and at follow-up evaluation. The association between baseline medication use and the subsequent development of IBD during follow-up evaluation was evaluated across 24 groups of medications (Table 3 and Supplementary Tables 4 and 5). Across the overall study population, angiotensin-converting enzyme inhibitors (5861; 4.4%), diuretics (5272; 4%), aspirin (5323; 4%), oral/injectable hypoglycemic medications (5232; 3.9%), and β-blockers (5027; 3.8%) were the most commonly used medications at baseline. In addition, approximately one third of females reported current or previous use of oral contraceptives at baseline.

A total of 947 (0.7%) participants reported antibiotic use at baseline, of which 16 developed IBD. Baseline antibiotic use was associated significantly with the development of IBD after minimal adjustment (OR, 3.10; 95% CI, 1.88–5.14; $P < .001$). After further adjustment for additional covariates, this finding remained significant (adjusted odds ratio [aOR], 2.81; 95% CI, 1.67–4.73; $P = .0001$). Results were consistent in direction for the outcomes of CD and UC individually, with no evidence of heterogeneity (tests of heterogeneity: chi-square; $P = .768$; $I^2 = 0.0\%$) (Figure 1). Similarly, results were directionally consistent among participants from high-/upper-middle–income countries and in those from low-/lower-middle–income countries, with no evidence of heterogeneity (tests of heterogeneity: chi-square; $P = .976$; $I^2 = 0.0\%$) (Supplementary Figure 1). Similar associations were observed with antituberculosis and antimalarial medications (OR, 6.41; 95% CI, 0.83–49.49; $P = .075$; and OR, 10.13; 95% CI, 1.38–74.49; $P = .023$, respectively), although these did not reach statistical significance.

Of the 24,610 (31.1%) females reporting previous or current oral contraceptive use at baseline, 185 developed IBD. Among female participants, previous or current oral contraceptive use at baseline also was associated significantly with the development of IBD after minimal adjustment (OR, 2.40; 95% CI, 1.90–3.04; $P < .001$), which remained significant after further adjustment (aOR, 2.17; 95% CI, 1.70–2.77; $P < .001$). Similarly, participants with baseline hormonal medication use (405; 0.3%) had an increased likelihood of developing IBD compared with those

What You Need to Know

Background

Medications such as nonsteroidal anti-inflammatory drugs (NSAIDs), antibiotics, and oral contraceptives are potential risk factors for inflammatory bowel disease (IBD).

Findings

This study found that antibiotics, oral contraceptives, and long-term NSAID use were associated with increased odds of incident IBD.

Implications for patient care

This study illuminates the importance of caution in prescribing antibiotics and oral contraceptives, and, if appropriate, discouraging patients from long-term use of NSAIDs to mitigate this established environmental risk of IBD.

Table 1. Characteristics of the Study Population at Time of Enrollment

	Overall (n = 133,137)	Participants who developed IBD (n = 571)	Participants who developed CD (n = 143)	Participants who developed UC (n = 428)
Cases of IBD, n (%) Crohn's disease, n (%) Ulcerative colitis, n (%)	571 (0.4) 143 (0.1) 428 (0.3)			
Age, *y*, SD	50.3 (9.7)	51.1 (9.4)	52.0 (9.4)	50.8 (9.4)
Female, n (%)	79,142 (59.5)	348 (61.0)	91 (63.6)	257 (60.1)
Urban, n (%)	69,147 (51.9)	362 (63.4)	103 (72.0)	259 (60.5)
Current smoker, n (%)	23,317 (17.7)	82 (14.4)	21 (14.8)	61 (14.3)
Education, n (%) None Secondary/high/higher secondary Trade or college/university	 55,896 (42.0) 49,948 (37.5) 26,722 (20.5)	 215 (37.7) 189 (36.2) 167 (32.0)	 69 (48.3) 35 (24.5) 39 (27.3)	 146 (34.1) 154 (36.0) 128 (29.9)
Country-specific income level High/upper-middle Low/lower-middle	 51,267 (38.5) 81,889 (61.5)	 310 (54.3) 261 (45.7)	 97 (67.8) 46 (32.2)	 213 (49.8) 215 (50.2)
BMI, means (SD)	25.9 (5.3)	26.4 (5.1)	26.3 (4.4)	26.5 (5.3)
Comorbid conditions, n (%)				
Diabetes	3969 (3.0)	9 (1.6)	1 (0.7)	8 (1.9)
Hypertension	11,895 (8.9)	21 (3.7)	3 (2.1)	18 (4.2)
Stroke	1149 (0.9)	0	0	0
Angina/heart attack/coronary artery disease	2888 (2.2)	6 (1.1)	1 (0.7)	5 (1.2)
Heart failure	392 (0.3)	0	0	0
Other heart disease	1168 (0.9)	2 (0.4)	0	2 (0.5)
Hepatitis/jaundice	981 (0.7)	2 (0.4)	0	2 (0.5)
Cancer	678 (0.5)	0	0	0
COPD	673 (0.5)	0	0	0
Asthma	1461 (1.1)	4 (0.7)	0	4 (0.9)
Tuberculosis	872 (0.7)	2 (0.4)	0	2 (0.5)

BMI, body mass index; CD, Crohn's disease; COPD, chronic obstructive pulmonary disease; IBD, inflammatory bowel disease; UC, ulcerative colitis.
*Among females.

who did not use hormonal medications after minimal adjustment (OR, 4.81; 95% CI, 1.97–11.77; $P = .001$), and this remained significant after further adjustment (aOR, 4.43; 95% CI, 1.78–11.01; $P = .001$). When the individual outcomes of CD and UC were evaluated, results were directionally consistent with no statistical evidence of heterogeneity for oral contraceptive use (tests of heterogeneity: chi-square; $P = .921$; $I^2 = 0.0\%$) and for hormonal medication use (tests of heterogeneity: chi-square; $P = .921$; $I^2 = 0.0\%$) (Figure 1). Results among participants from high-/upper-middle–income countries and those from low-/lower-middle–income countries also were consistent in direction with no evidence of heterogeneity for oral contraceptive use (tests of heterogeneity: chi-square; $P = .878$; $I^2 = 0.0\%$) and for hormonal medication use (tests of heterogeneity: chi-square; $P = .992$; $I^2 = 0.0\%$) (Supplementary Figure 1).

A total of 2939 (2.2%) participants reported NSAID use at baseline, of whom 29 developed IBD. NSAID use at baseline also was observed to have a significant association with the development of IBD (OR, 1.81; 95% CI, 1.24–2.65; $P = .002$), which remained significant after adjustment for additional covariates (aOR, 1.80; 95% CI, 1.23–2.64; $P = .002$). Results were directionally consistent in CD and for UC with no statistical evidence of heterogeneity (tests of heterogeneity: chi-square; $P = .905$; $I^2 = 0.0\%$) (Figure 1). Similarly, results were consistent in direction among participants from high-/upper-middle–income countries and those from low-/lower-middle–income countries with no

Table 2. Medication Use Among the Study Population

Medication, n (%)	Short-term (baseline use)		Long-term (baseline and follow-up evaluation)	
	Overall (n = 133,137)	Participants who developed IBD (n = 571)	Overall (n = 133,137)	Participants who developed IBD (n = 571)
Angiotensin-converting enzyme inhibitors	5861 (4.4)	33 (5.8)	2656 (2.0)	21 (3.7)
β-blockers	5027 (3.8)	37 (6.5)	2269 (1.7)	14 (2.5)
Calcium antagonists	4660 (3.5)	9 (1.6)	1549 (1.2)	5 (0.9)
Diuretics	5272 (4.0)	28 (4.9)	1531 (1.1)	14 (2.5)
Statins	4611 (3.5)	47 (8.2)	2324 (1.7)	25 (4.4)
Aspirin	5323 (4.0)	46 (8.1)	1985 (1.5)	18 (3.2)
Clopidogrel	301 (0.2)	2 (0.4)	43 (0.1)	0
Vitamin K antagonists	587 (0.4)	2 (0.4)	123 (0.1)	2 (0.4)
Oral/injectable hypoglycemic medications	5232 (3.9)	4 (0.7)	2343 (1.8)	14 (2.5)
Nitrates	660 (0.5)	7 (1.2)	121 (0.1)	1 (0.2)
α antagonists	367 (0.3)	2 (0.4)	19 (0.1)	0
Angiotensin II receptor blockers	2424 (1.8)	24 (4.2)	968 (0.7)	10 (1.8)
Antituberculosis medications	63 (0.1)	1 (0.2)	1 (0.1)	0
Antiarrhythmic medications	189 (0.1)	5 (0.9)	48 (0.1)	0
Blood thinners	747 (0.6)	2 (0.4)	80 (0.1)	0
NSAIDs	2939 (2.2)	29 (5.1)	147 (0.1)	5 (0.9)
Antibiotics	947 (0.7)	16 (2.8)	25 (0.1)	0
Oral contraceptives[a]	24,610/79,142 (31.1)	185/348 (53.2)		
Proton pump inhibitors	294 (0.2)	1 (0.2)		
Antimalarial medications	68 (0.1)	2 (0.4)	1 (0.1)	0
Anticonvulsant medications	99 (0.1)	1 (0.2)	1 (0.1)	0
Antihistamine medications	104 (0.1)	1 (0.2)	0	0
Hormonal medications	405 (0.3)	5 (0.9)	0	0
Antiviral medications	47 (0.1)	1 (0.2)		

IBD, inflammatory bowel disease; NSAID, nonsteroidal anti-inflammatory drug.
[a]Among females.

evidence of heterogeneity (tests of heterogeneity: chi-square; $P = .544$; $I^2 = 0.0\%$) (Supplementary Figure 1).

Long-Term Medication Use and the Development of Inflammatory Bowel Disease

Sensitivity analyses evaluated whether long-term medication use (defined as use at baseline and follow-up evaluation) had any impact on the outcomes of interest. Medications with a significant association with the development of IBD were evaluated in this analysis. Overall, among the 947 participants with antibiotic use at baseline, 25 participants continued to use antibiotics during follow-up evaluation. None of the 16 participants with baseline antibiotic use who developed IBD had long-term antibiotic use. Furthermore, none of the participants with baseline hormonal medication use who subsequently developed IBD had long-term use. Participants with long-term NSAID use had a greater likelihood of developing IBD when fully adjusted for covariates (aOR, 5.58; 95% CI, 2.26–13.80; $P < .001$). All participants with long-term NSAID use were from high-/upper-middle–income countries and tests for heterogeneity to compare against lower-income countries were not performed. Results were consistent in direction for CD and UC, with no evidence of statistical heterogeneity (tests of heterogeneity: chi-square; $P = .999$; $I^2 = 0.0\%$) (Figure 2). Short-term users of NSAIDS (use at baseline

Table 3. Association Between Baseline Medication Use and Development of IBD

	Unadjusted		Minimally adjusted		Fully adjusted	
Medication	OR (95% CI)	*P* value	OR (95% CI)	*P* value	OR (95% CI)	*P* value
Angiotensin-converting enzyme inhibitors	1.33 (0.94–1.90)	.109	1.01 (0.70–1.44)	.969	1.00 (0.67–1.49)	.985
β-blockers	1.77 (1.27–2.47)	<.001	1.25 (0.89–1.75)	.200	1.30 (0.90–1.88)	.163
Calcium antagonists	0.44 (0.23–0.85)	.015	0.40 (0.21–0.78)	.007	0.35 (0.16–0.75)	.006
Diuretics	1.25 (0.86–1.83)	.247	1.20 (0.82–1.77)	.351	1.30 (0.86–1.98)	.2178
Statins	2.52 (1.86–3.40)	1.65E-09	1.70 (1.25–2.32)	.001	1.44 (1.01–2.04)	.043
Aspirin	2.11 (1.56–2.86)	1.26E-06	1.65 (1.21–2.25)	.002	1.65 (1.18–2.29)	.003
Clopidogrel	1.55 (0.39–6.26)	.535	0.90 (0.22–3.61)	.876	0.46 (0.06–3.27)	.436
Vitamin K antagonists	0.79 (0.20–3.19)	.744	0.39 (0.10–1.57)	.187	0.42 (0.10–1.70)	.224
Oral/injectable hypoglycemic medications	1.41 (0.98–2.02)	.066	1.13 (0.78–1.63)	.516	0.97 (0.63–1.50)	.886
Nitrates	2.51 (1.19–5.30)	.016	2.41 (1.13–5.12)	.022	2.84 (1.31–6.13)	.008
α antagonists	1.27 (0.32–5.12)	.734	0.78 (0.19–3.15)	.727	0.82 (0.20–3.34)	.785
Angiotensin II receptor blockers	2.38 (1.58–3.59)	3.52E-05	1.43 (0.94–2.17)	.091	1.24 (0.77–1.98)	.371
Antituberculosis medications	4.06 (0.56–29.34)	.165	6.30 (0.87–45.89)	.069	6.41 (0.83–49.49)	.075
Antiarrhythmic medications	6.36 (2.60–15.51)	4.85E-05	4.67 (1.90–11.44)	.001	3.48 (1.10–11.05)	.034
Blood thinners	0.62 (0.15–2.50)	.503	1.01 (0.25–4.08)	.986	0.62 (0.09–4.45)	.634
NSAIDs	2.59 (1.78–3.78)	6.87E-07	1.81 (1.24–2.65)	.002	1.80 (1.23–2.64)	.002
Antibiotics	4.43 (2.68–7.31)	6.26E-09	3.10 (1.88–5.14)	1.05E-05	2.81 (1.67–4.73)	.0001
Oral contraceptives[a]	2.82 (2.40–3.56)	2.07E-18	2.40 (1.90–3.04)	2.75E-13	2.17 (1.70–2.77)	5.02E-10
Proton pump inhibitors	0.79 (0.11–5.65)	.816	1.40 (0.20–10.06)	.735	1.41 (0.20–10.24)	.732
Antimalarial medications	7.06 (1.72–28.88)	.007	13.92 (3.38–57.31)	.001	10.13 (1.38–74.49)	.023
Anticonvulsant medications	2.37 (0.33–17.03)	.391	4.31 (0.60–31.05)	.147	4.68 (0.64–34.36)	.130
Antihistamine medications	2.26 (0.31–16.20)	.418	4.44 (0.62–31.98)	.139	4.26 (0.57–31.71)	.157
Hormonal medications	2.92 (1.20–7.08)	.018	4.81 (1.97–11.77)	.001	4.43 (1.78–11.01)	.001
Antiviral medications	5.06 (0.70–36.71)	.109	9.82 (1.34–71.64)	.024	11.96 (1.59–90.17)	.016

NOTE. Minimally adjusted for age, sex, and center. Fully adjusted for age, sex, center, smoking, body mass index, education, location, Alternative Healthy Eating Index score, diabetes, hypertension, stroke, angina/heart attack/coronary artery disease, heart failure, other heart disease, hepatitis/jaundice, cancer, chronic obstructive pulmonary disease, asthma, and tuberculosis.
CI, confidence interval; IBD, inflammatory bowel disease; NSAID, nonsteroidal anti-inflammatory drug; OR, odds ratio.
[a]Among 79,142 females.

but not at follow-up evaluation) had no significant association with the development of IBD. Again, results were directionally consistent with CD and UC, with no evidence of heterogeneity (tests of heterogeneity: chi-square; $P = .901$; $I^2 = 0.0\%$). Results among participants from high-/upper-middle–income countries and those from low-/lower-middle–income countries were similar in direction with no evidence of substantial heterogeneity (tests of heterogeneity: chi-square; $P = .208$; $I^2 = 36.8\%$) (Supplementary Figure 2).

Discussion

In this large prospective cohort study of 133,137 individuals from 24 low-, middle-, and high-income countries, we found multiple associations between medication use and odds of IBD. Use of antibiotics and hormonal therapies including oral contraceptive pills were associated with an increased odds of IBD, as was longer-term use of NSAIDs. For positive associations identified in our study, similar directional effects were seen for both UC and CD, with no significant heterogeneity. We were unable to confirm or identify any other associations that previously have been reported in the literature, including with proton pump inhibitor or statin use.[11,12] Our analysis illuminates and expands upon the role that medications may play as an environmental factor associated with risk of IBD.

The pathogenesis of IBD is poorly understood, however, the complex interactions between genetic

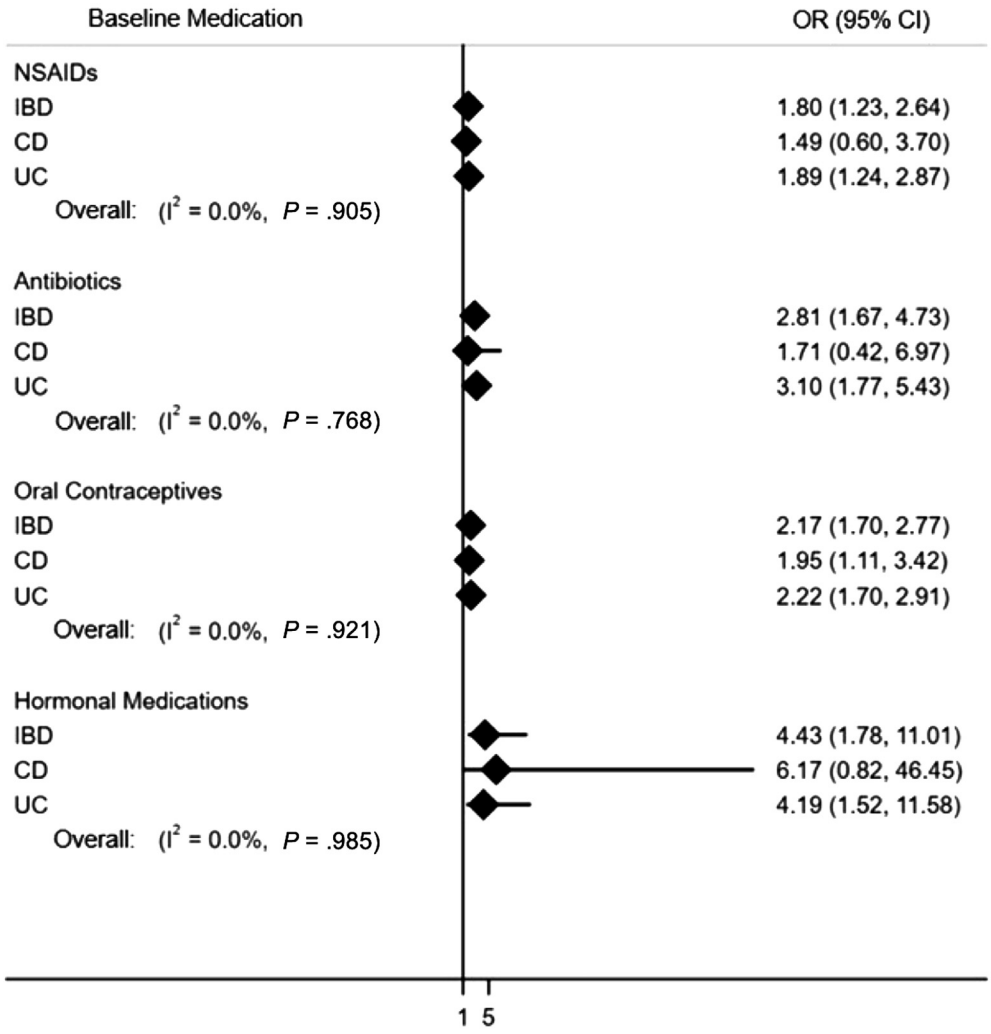


Figure 1. Forest plot of baseline medication use by IBD, CD, and UC. CD, Crohn's disease; IBD, inflammatory bowel disease; OR, odds ratio; UC, ulcerative colitis.

predisposition and environmental factors increasingly have been recognized. More than 100 genes have been identified and are implicated in rendering individuals susceptible to IBD, although genetic factors alone are insufficient to account for the increasing prevalence of IBD in developed and developing countries, and the sharp increase of incidence in developing nations.[2,16,17] Environmental determinants of IBD increasingly are being recognized, including the role that medications play in altering the intestinal environment, mucosal barrier, and gut microbiome, and the associated increased risk these alterations pose in the development of IBD.[17] The global use of medications (both with and without a prescription) has been increasing over the past 2 decades, with a current estimated 1 dose per person per day of medicines in more than half of the world's population, a 33% increase compared with 2005.[18] The largest increase of medication use aligns with regions with some of the largest developmental gains including India, China, Brazil, Indonesia, and Africa, areas that previously had some of the lowest use of medications.[18] Given the changing epidemiological pattern of IBD and global medication use, there is a need to better understand medications as potential environmental risk factors for the development of IBD.[17]

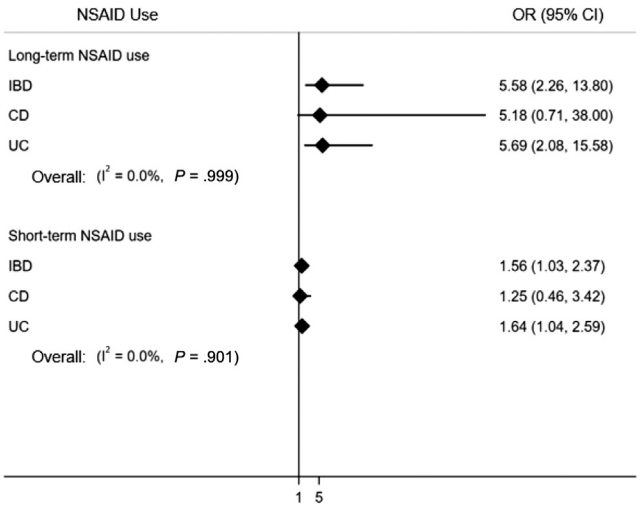


Figure 2. Forest plot of long-term and short-term NSAID use by IBD, CD, and UC. CD, Crohn's disease; IBD, inflammatory bowel disease; NSAID, nonsteroidal anti-inflammatory drug; OR, odds ratio; UC, ulcerative colitis.

Nonsteroidal Anti-Inflammatory Drugs

Our analysis showed an association between baseline NSAID use and increased odds of IBD, mainly observed with longer-term use of NSAIDs. Aspirin was recorded separately and also had a similar trend observed for increased odds (aOR, 1.65; 95% CI, 1.18–2.29; $P = .003$). Prolonged NSAID use contributes to the change and destruction of the intestinal barrier that may predispose individuals to developing IBD. This likely is multifactorial including topical effects, reduced prostaglandins, and microbial dysbiosis, which, ultimately, alters gut permeability, motility, and generates oxidative species that together affect and potentially exacerbate inflammatory responses.[19–22] Previously published literature from prospective cohort studies similarly has shown increased risk of IBD with longer-term NSAID use[21] and increased risk of CD in patients with regular aspirin use.[23] Moreover, patients with IBD who regularly use NSAIDs have a higher risk of clinically active disease and higher rates of disease relapse.[24,25]

Antibiotics

We also observed that baseline use of antibiotics was associated with increased odds of IBD. The effect of antibiotics on the intestinal environment and relationship with IBD has been investigated extensively. Antibiotic exposure in healthy individuals has been shown to alter the gut microbiota, with increased frequency and duration of antibiotic exposure resulting in persistent dysbiosis.[26–28] These changes in gut microbial diversity and loss of protective microbial species secondary to antibiotic exposure has been shown previously to be an environmental risk factor of IBD in a dose-dependent manner; however, robust and consistent data are lacking.[29] This inconsistency likely is owing to the heterogeneity of past studies including retrospective and case-control analyses and the inherent limitations of these study designs. Moreover, different classes of antibiotics affect gut microbiome differently. Previous studies have reported differing associations between antibiotic classes and IBD, such as the consistently increased risk with metronidazole compared with other classes.[30,31] Although our study did not permit further subgroup analysis by antibiotic classes and we were unable to determine the indication for antibiotic prescription, our study has the benefit of a large and diverse population analyzed prospectively.

Oral Contraceptives and Hormonal Medications

Hormonal medications used by males or females at baseline and prior/current oral contraceptive use in women at baseline also was associated with increased odds of IBD. Previous literature has shown divergent results with regard to the associations between oral contraceptive pills and IBD. Notably, a strong association between contraceptive pills and CD has been shown in multiple analyses, although the relationship with UC has not been observed consistently.[32–34] In postmenopausal women who use hormonal medications, a relationship between increased risk of UC, but not CD, has been reported.[32] The precise mechanism underlying the hormonal relationship and risk of IBD and effect on gastrointestinal environment is poorly understood, although it has been proposed that estrogen may affect the humoral immune system and modify the colonic barrier.[35,36]

Strengths and Limitations

Our study had several strengths. We used a large multinational cohort and prospectively investigated associations between medication use and incident IBD. This study design overcomes the limitations of many existing studies within this field because most were retrospective, case-control designs or used homogenous populations, which limits external validity. However, our study had a few potential limitations. The bulk of the population within PURE were adults aged 35 to 70, and it is unclear if these associations apply to younger individuals, although other cohort studies have shown similar associations among participants of younger ages.[21,32,37] There are far more UC diagnoses compared with CD, which likely reflects the bimodal nature of a UC diagnosis, with many individuals being diagnosed in their 50s and 60s, in comparison with the lower incidence of CD in older individuals.[38] The lower number of CD cases observed in our cohort may have resulted in insufficient power to detect some associations of medications with IBD risk. Because follow-up visits occurred every year, there is a potential for misclassification of exposure over time because it is unlikely that all medications assessed in this study were used consistently throughout the follow-up period. In addition, participants self-reported CD and UC diagnoses, and medical records in English were selected randomly to validate diagnoses. Because only a sample of English language medical records underwent verification it is possible that some cases of IBD were misclassified. Although this may introduce bias, misclassification would more likely bias associations to the null and not affect results. Furthermore, medication use was collected using a survey that gathered information on whether a participant was taking a medication at least once per week. No data were gathered with regard to dose and duration at study enrollment, or before enrollment. All medication use was self-reported by participants, which is subject to recall error and potential unwillingness to disclose the use of certain medications. With regard to oral contraceptive pills, we were unable to separate current use from

previous use. Data regarding oral contraceptive use during follow-up evaluation was not recorded and we were unable to evaluate the impact of long-term use. Age at menopause onset was unavailable in PURE and we were unable to assess the association between oral contraceptive use among premenopausal women specifically. In addition, information on medication subclasses within groups (ie, different types of antibiotics) was not collected. Further strengths and limitations are discussed in the Supplementary Methods section.

Conclusions

As the global burden of IBD increases, it is imperative to identify environmental determinants that may be contributing to this trend. In this study, antibiotics, hormonal medications, oral contraceptives, and prolonged use of NSAIDs were associated with a higher risk of IBD. Additional studies are warranted to further elucidate precise risk profiles of medications we identified, such as duration of therapy, dose, and specific classes of medications that confer an increased risk of IBD. Nevertheless, this study illuminates the importance of caution in prescribing antibiotics and oral contraceptives, and, if appropriate, discouraging patients from long-term use of NSAIDs to mitigate this established environmental risk of IBD.

Supplementary Material

Note: To access the supplementary material accompanying this article, visit the online version of *Clinical Gastroenterology and Hepatology* at www.cghjournal.org, and at http://doi.org/10.1016/j.cgh.2022.11.037.

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Conflicts of interest

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Supplementary Methods

Details Regarding Study Design

In short, the first and second phases of the PURE study occurred between January 1, 2003, and December 31, 2016. A total of 133,137 participants between ages 20 and 80 years with available data from at least 1 follow-up evaluation were included in this study. Participants from 24 countries were included (Argentina, Bangladesh, Brazil, Canada, Chile, China, Colombia, India, Iran, Kazakhstan, Kyrgyzstan, Malaysia, Pakistan, Palestine, Philippines, Poland, Russia, Saudi Arabia, South Africa, Sweden, Tanzania, Turkey, United Arab Emirates, and Zimbabwe). IBD prevalence in 1990 and 2017 for both sexes and all countries included is provided in Supplementary Table 6. The focus of recruitment in PURE was adults between ages 35 and 70 years, although some participants younger than age 35 (n = 547) or older than age 70 years of age (n = 369) were recruited, the majority from China (n = 308) and Malaysia (n = 239). A detailed sampling and recruitment strategy of the PURE study is provided later. Briefly, standardized questionnaires were used to collect participant data at baseline and follow-up visits, which occurred at least every 3 years. Data were collected at the community, household, and individual level. For example, data regarding tobacco use was measured at the community level using tobacco pricing and availability, at the household level using the number of tobacco users per household, and at the individual level using tobacco exposure. New diagnoses of CD and UC were recorded on case report forms during follow-up evaluation. The current analysis considered all outcome events up to April 2, 2021. The PURE study is coordinated by the Population Health Research Institute, McMaster University, and Hamilton Health Sciences in Hamilton, Ontario, Canada. The PURE study is registered on ClinicalTrials.gov, identifier: NCT03225586. The Hamilton Integrated Research Ethics Board has approved of the conduct of this study through protocol #03-206.

Variables

During the baseline and follow-up visits, participants were asked if they were taking any medications regularly (defined as at least once per week in the past month) and to list any current medications they were consuming. Additional information such as medication dose, frequency, duration of use, and route of administration were not asked or recorded in the PURE study. Medications were coded using standardized methods and grouped into medication classes. A total of 24 medication groups were evaluated, including angiotensin-converting enzyme inhibitors, β-blockers, calcium antagonists, diuretics, statins, aspirin, clopidogrel, vitamin K antagonists, oral/injectable hypoglycemic medications, nitrates, α antagonists, angiotensin II receptor blockers, antituberculosis medications, antiarrhythmic medications, blood thinners, NSAIDs, antibiotics, oral contraceptives, proton pump inhibitors, antimalarial medications, anticonvulsant medications, antihistamine medications, hormonal medications, and antiviral medications. Oral contraceptives were evaluated only among participants who identified as female. At baseline, participants were asked about previous or current oral contraceptive use. For all other medications, participants reported current use only. A full list of medications is detailed in Supplementary Table 1. However, data regarding specific medications by generic name were not available at the participant level. Medications were assessed as dichotomous variables. Baseline use was defined as medication use reported at enrollment, regardless of follow-up use. Short-term was defined as medication use at baseline but not at follow-up evaluation. Long-term was defined as medication use reported at baseline and at next follow-up visit (approximately 3 years apart).

At each follow-up visit, participants were asked if they were diagnosed with CD or UC and when the diagnosis occurred. Participants with an IBD diagnosis before the baseline questionnaire were excluded. Given the self-reported nature of the diagnoses, a sample of the population with medical records that could be reviewed by the authors was used to validate 20% of the diagnoses, which is detailed in the Supplementary Appendix of a previous publication.[1] Further subgroup analyses were planned to evaluate significant associations by income level and tests for heterogeneity were performed. The income level of a country was based on the gross domestic product per capita. High-income countries included Canada, Sweden, United Arab Emirates, and Saudi Arabia. Upper-middle–income countries included Argentina, Brazil, Chile, Malaysia, Poland, South Africa, Turkey, Russia, and Kazakhstan. Lower-middle–income countries included China, Philippines, Colombia, Iran, Palestine, and Kyrgyzstan. Low-income countries included Bangladesh, India, Pakistan, Zimbabwe, and Tanzania. Income level was adjusted for because it previously showed an association with prescription medication use in IBD.[2]

Statistical Analysis

Age and body mass index were treated as continuous variables. Categoric variables included participant center, smoking (current/former/never), and education, categorized as college/university/trade (>11 years of age), secondary/high school (7–11 years of age), or none/primary (first 6 years of age). Binary variables included sex, location (rural/urban), and existence of comorbid conditions, including diabetes, hypertension, stroke, angina/heart attack/coronary artery disease, heart failure, other heart disease, hepatitis/jaundice, cancer, chronic obstructive pulmonary disease, asthma, and tuberculosis.

To determine whether associations were owing primarily to a subgroup effect, heterogeneity of the

subgroup analyses was assessed by calculating chi-square and I^2 statistics. I^2 values that were greater than 50% indicated substantial heterogeneity. A *P* value less than .10 was used to indicate substantial heterogeneity for the chi-square test. To account for multiple hypotheses testing, a Bonferroni correction was applied to the conventional *P* value threshold. For analyses evaluating baseline medication use, a total of 24 medications were assessed, corresponding to a 2-sided *P* value threshold of .0021. This reduced threshold also was used for subanalyses of short-term vs long-term use and for CD vs UC. Data were analyzed using Stata/IC version 15.0 (StataCorp LLC).

Selection Methodology

PURE study participant selection was as described from Teo et al.[3]

Selection of countries

The choice and number of countries selected in PURE reflects a balance between involving a large number of communities in countries at different economic levels, with substantial heterogeneity in social and economic circumstances and policies, and the feasibility of centers to successfully achieve long-term follow-up evaluation. Thus, PURE included sites in which investigators are committed to collecting good-quality data for a low-budget study over the planned 10-year follow-up period and did not aim for a strict proportionate sampling of the entire world.

Selection of communities

Within each country, urban and rural communities were selected based on broad guidelines. A common definition for *community* that is applicable globally is difficult to establish. In PURE, a community was defined as a group of people who have common characteristics and reside in a defined geographic area. A city or large town was not usually considered to be a single community, rather communities from low-, middle-, and high-income areas were selected from sections of the city and the community area was defined according to a geographic measure (eg, a set of contiguous postal code areas or a group of streets or a village). The primary sampling unit for rural areas in many countries was the village. The reason for inclusion of both urban and rural communities is that, for many countries, urban and rural

Environments show distinct characteristics in social and physical environments, and, hence, by sampling both, we ensured considerable variation in societal factors across PURE communities. The number of communities selected in each country varied, with the aim to recruit communities with substantial heterogeneity in social and economic circumstances balanced against the capacity of local investigators to maintain follow-up evaluation. In some countries (eg, India, China, Canada, and Colombia), communities from several states/provinces were included to capture regional diversity, in policy, socioeconomic status, culture, and physical environment. In other countries (eg, Iran, Poland, Sweden, and Zimbabwe), fewer communities were selected.

Selections of households and individuals

Within each community, sampling was designed to achieve a broadly representative sample of that community of adults aged between 35 and 70 years. The choice of sampling frame within each center was based on both representativeness and feasibility of long-term follow-up evaluation, following broad study guidelines. Once a community was identified, where possible, common and standardized approaches were applied to the enumeration of households, identification of individuals, recruitment procedures, and data collection. The method of approaching households differed between regions. For example, in rural areas of India and China, a community announcement was made to the village through contact of a community leader, followed by in-person, door-to-door visits of all households. In contrast, in Canada, initial contact was by mail, followed by telephone, inviting members of the households to a central clinic. Households were eligible if at least 1 member of the household was between the ages of 35 and 70 years and the household members intended to continue living in their current home for a further 4 years. For each approach, at least 3 attempts at contact were made. All individuals within these households between ages 35 and 70 years providing written informed consent were enrolled. When an eligible household or eligible individual in a household refused to participate, demographics and self-reported data about cardiovascular disease risk factors, education, and history of cardiovascular disease, cancers, and deaths in the households within the 2 previous years were recorded. To ensure standardization and high data quality, we used a comprehensive operations manual, training workshops, DVDs, regular communication with study personnel, and standardized report forms. We entered all data in a customized database programmed with range and consistency checks, which was transmitted electronically to the Population Health Research Institute in Hamilton (Ontario, Canada) where further quality checks were implemented.

Additional Strengths and Limitations of the Current Study

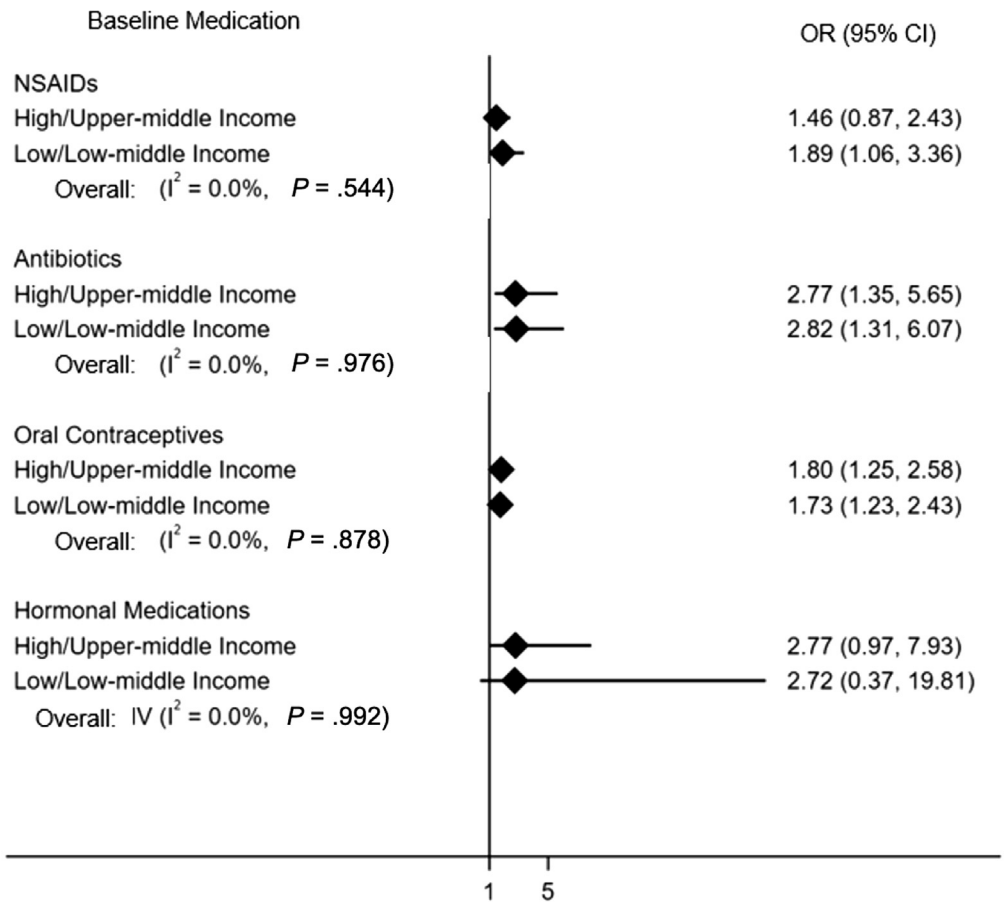
Although randomized controlled trials often are thought to be the optimal study design for minimizing bias, the relatively low incidence of IBD even in regions where it is more common and the inherent difficulties

that come with assessing harm in randomized controlled trials make prospective cohort studies (such as PURE) the most pragmatic reliable study design to assess for environmental risks associated with IBD. Medication use was documented at baseline, before IBD diagnosis, which minimizes the risk of recall bias. Our analyses were adjusted for potential confounding factors that also were associated with risk of IBD. We also adjusted for multiplicity of groups of drugs evaluated. Country-specific questionnaires that were validated and standardized were used in collection of baseline and follow-up medication use. The similar patterns of increased risk observed in this study for NSAIDs and aspirin, and for oral contraceptives and hormonal therapies, provide increased confidence in the results owing to the consistency of these findings. Similarly increased risks also were observed within higher- and lower-income countries with no significant heterogeneity. The increased risk of IBD with NSAIDs, oral contraceptives, hormonal therapies, and antibiotics have been shown previously in separate studies, and we show these associations in a single international prospective study with lower-income countries. Furthermore, we adjusted for a broad range of covariates previously identified to have an association with the development of IBD in the PURE cohort, including diet. Lastly, our results satisfy most of Hill's[4] criteria for assessing causation between a presumed cause and an observed effect, except for establishment of a biologic gradient, because the dose–effect relationship could not be examined because of a lack of data collected around dosing.

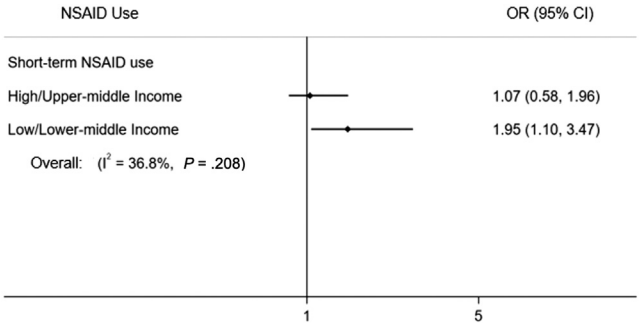
Our definition of long-term medication use included those who reported use at baseline and at follow-up evaluation, although it is possible that the medication was not used consistently, or not used at all, between these time points. In particular, there were only 5 cases with long-term NSAID use, which were all from high-/upper-middle–income countries, which precludes any definitive conclusions. Furthermore, there is a potential for immortal time bias in our long-term medication use analyses because cases diagnosed between baseline and follow-up evaluation were not associated with long-term use. Similarly, short-term medication use was defined as use at baseline, but not at follow-up evaluation. It remains plausible that individuals were using a medication for a prolonged period of time (months or years), but ultimately were considered a short-term user in our analysis, which may be a potential source of misclassification bias in our analysis. Furthermore, it is conceivable that some results could be false positives as a result of multiple comparisons, however, we used Bonferroni correction to account for multiple testing and the reported associations met the reduced threshold for statistical significance. Lastly, using multivariable logistic regression models, we adjusted for many variables, however, because this study was observational in nature it is possible that there is residual bias from unmeasured, imprecise variables or unknown confounders. For example, the imprecision with which smoking status was categorized (current, former, or never) could be a source of residual confounding.

Supplementary References

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3. Teo K, Lear S, Islam S, et al. Prevalence of a healthy lifestyle among individuals with cardiovascular disease in high-, middle- and low-income countries: the Prospective Urban Rural Epidemiology (PURE) study. JAMA 2013;309:1613–1621.
4. Hill AB. The environment and disease: association or causation? Proc R Soc Med 1965;58:295–300.



Supplementary Figure 1. Forest plot of baseline medication use by income level. NSAID, nonsteroidal anti-inflammatory drug; OR, odds ratio.



Supplementary Figure 2. Forest plot of short-term nonsteroidal anti-inflammatory drug (NSAID) use by income level. OR, odds ratio.

Supplementary Table 1. List of Medications by Generic Name

Medication group	Generic names
Angiotensin-converting enzyme inhibitors	Captopril (Aceten, Apuzin, Capoten, Captoril, Epsitron) Enalapril (Anapril, Angiocare, Enace, Enacard, Enaladex, Enalten, Renitec, Vasotec) Fosinopril (Fovas, Fosinace, Monopril) Lisinopril (Cipril, Lame 10, Prinivil, Zestril) Perindopril (Aceon, Coversyl, Coversum) Ramipril (Altace, Ramace, Tritace) Benazepril (Benace, Lotensin) Trandolapril (Mavik, Zetpril) Quinapril (Accupril, Acuitel, Quinaprilum) Zofenopril/Zofenil (Bifril, Zocardis, Zofecard, Zopranol, Zantipres) Cilazapril (Apo-Cilazapril, Aceprix, Cilan, Cazaprol, Dynorm, Inhibace, Vascase) Imidapril (Hipertene, I-Prill, Imidagi, Tanatril) Moexipril (Univasc)
β-blockers	Atenolol (Aten, Atecard, Altol, Atelol, Atenex, Atenopress, Atenova, Atpark, Normitab, Ratio-Tenolin, Tenormin, Tenolol, Vascoten, Ziblok) Bisoprolol (Bisocor, Cardicor, Monocor, Niperten, Zebeta, Concor) Carvedilol (Aricard, Carloc, Carveda, Coreg, Karvil) Esmolol (Brevibloc) Labetalol (Gravidol, Trandate, Normodyne) Metoprolol (Betaloc, Denex, Lopressor, Mepresor, Seloken, Toprol-XL) Propanalol/Propanolol/Propranolol (Avlocardyl, Ciplar, Deralin, Inderal, InnoPran XL) Nebivolol (Nebilet, Bystolic, Nebicard, Nebicip) Timolol (Blocadren) Sotalol (Betapace, Betapace AF, Sorine, Solet, Sotagard) Acebutolol (Sectral) Nadolol (Apo Nadol, Corgard) Pindolol (Visken, Apo-Pindolol) Betaxolol (Kerlone, Lokren) Alprenolol/Alfeprol/Alpheprol (Regletin, Gubernal) Celiprolol (Celipres, Celiprol, Dilanorm, Selectol) Bupranolol (Adomed)
Calcium antagonists	Amlodipine (Amcard, Amcor, Amlip, Amlocor, Amlomed, Amlokind, Amlor, Stamlo, Norvasc) Diltiazem (Angizem, Cardizem, Cartia XT, Dilacor XR, Dilzem, Diltzac, Dilt-CD, Herbesser, Tiazac) Felodipine (Plendil, Prevex) Nifedipine (Adalat, Adalat XL, Calcigard, Procardia, Procardia XL, Nifedical XL, Zanifed) Lercanidipine (Cornelin, Lervasc, Landip-10, Lerka, Lerkanidipin, Lerez, Zanidip, Zanipin) Nimodipine (Nimotop, Nymalize) Verapamil (Calan, Calan SR, Covera-HS, Isoptin, Isoptin SR, Verelan, Vasopten) Cilnidipine (Atelec, Cilacar) Lacidipine (Lacipil, Lacivas, Lacibloc, Lacicard, Lacidac, Motens, Sinopil) Manidipine (Caldine, Iperten, Manivasc) Benidipine (Benipin, Codipine, Caritec, Coniel) Barnidipine/Mepirodipine (Barnix, Cyress, HypoCa) Isradipine (DynaCirc, DynaCirc CR, Prescal) Levamlodipine/S-Amlodipine (Asomex, Esam, S-Amcard) Nitrendipine (Baypress, Cardif, Nitrepin, Nitrepress, Nitrencord) Bepridil (Vascor)
Diuretics	Amiloride (Midamor) Hydrochlorothiazide (Apo-Hydro, Aquazide H, Esidrix, Microzide, Novo-Hydrazide, Oretic, Zide) Chlorthalidone (Hygroton, Thalitone) Indapamide/Metindamide (Arifon, Indamox, Ipamix, Lozide, Lozol, Natrilix, Natrix, Preterax, Tandix) Frusemide/Furosemide (Diaqua-2, Furocot, Frusenex, Fulsix, Lasix, Lo-Aqua) Spironolactone (Aldactone, Carospir, Novo-semide, Spiractin, Spiroton) Eplerenone (Inspra, Suficard) Torasemide/Torsemide (Demadex, Tortas) Metolazone (Diulo, Metenix, Mykrox, Zaroxolyn, Zytanix) Acetazolamide (Diamox, Diamox Sequels) Mannitol (Osmitrol, Resectisol) Triamterene/Teridin (Dyrenium, Dytac, Riyazine) Ethacrynic Acid (Edecrin) Polythiazide (Renese)

Supplementary Table 1. Continued

Medication group	Generic names
	Benzthiazide/Benzothiazide (Aquatag, Exna) Bendroflumethiazide (Aprinox, Centyl, Naturetin)
Statins	Atorvastatin (Atopil, Aplacase, Arpitor, Atorval, Storvas, Lipirex, Lipitor, Lipitor Stat-A) Lovastatin (Altoprev, Elstatin, Lestric, Lostatin, Mevacor, Rizonet, Sancos, Sunocor) Pravastatin (Pravator, Pravachol) Rosuvastatin (Crestor, Cougar, Ezallor, Loster, Prevast, Rosucard, Rozucard) Simvastatin (Simcard, Viscor, Zimstat, Zocor) Fluvastatin (lescol, Lescol-XL)
Aspirin	Acetylsalicylic acid (Adiro, Cardiomagnyl, aspirin, Ecosprin, Ecotrin, Loprin, baby aspirin, low-dose aspirin)
Clopidogrel	Clopidogrel (Aclotil, Agranil, Anplat, Aptogrel, Baclan, Cloflow, Plavix, Tromborel, Zyllt)
Vitamin K antagonists	Warfarin (Coumadin, Esberiven Depot, Jantoven, Marevan, Warf, Warfen, Warin) Phenindione (Dindevan, Fenindion, Phindivan) Dicoumarol Acenocoumarol (Acitrom, Nistrom, Sintrom) Coumarin + Troxerutin (Venalot)
Oral/injectable hypoglycemic medications	α-glucosidase inhibitors Acarbose (Dibose, Glucobay, Prandase, Precose, Sugatrol) Miglitol (Diaban, Glyset, Elitox) Voglibose (Advog, Oglibo, Vobose, Vobit, Voglinorm, Volibo, Volix, Vozuca, Voglib) Sulfonylureas Chlorpropamide (Copamide, Diabinese) Glibenclamide/Glyburide (Allase, Aviglen, Benformin, Claben, Codica, Coranil, Daonil, Daosin, D-Con 5, Declamide, Diabeta, Diabetnil, Diacare, Diatab, Euglucon, Gardoton, Getrol, Gilemal, Glibil, Glidanil, Glikel, Glimel, Glionil, Glucon, Glunil, Glybovin, Glycron, Glynase, PMS Glyburide, Ratio-glyburide, Maninil, Micronase, Novo-glyburide, NovoSemi, Semi-Daonil) Acetohexamide (Dimelin, Dymelor, Gamadiabet) Tolbutamide (Diaval, Orinase, Rastinon) Gliclazide (Betanorm, Comprid, Diabend, Diamicron, Diaprel MR 60, Diatrol, Efikas, Glaze, Glicasil, Glidex, Glizid, Glyade MR, Glycor, Glysigon, Gored, Nuzide, Reclide) Glimepiride (Amaryl, Amiran, Azulix, Bioglic, Diameprid, Evopride, Euglym, Getryl, Glemaz, Glemep, Glibetic, Glimaryl, Glimer, Glimpid, Glorion, Glucorid, Glypin, Zoryl) Glipizide (Diaplus, Dibizide, Glez, Glimerol, Glucotrol, Glucotrol XL, Glyzid, Semi Glynase) Biguanides Phenformin (Fenformin, DBI Ciba-Geigy, DBITD) Thiazolidinediones Rosiglitazone (Avandia, Enselin, Rezult, Rosenda, Rosicon) Pioglitazone (Actos, Dropia, Glifix, Glitazone, G-Tase 30, Pio 15, Pioglar, Pioglit, Piogtan, Piosafe, Pioz, Piozer, PMS-pioglitazone, Sandoz Pioglitazone, Zolid) Troglitazone (Rezulin, Romozin, Noscal) Dipeptidyl peptidase-4 inhibitors Sitagliptin (Januvia, Tesavel, Xelevia) Vildagliptin (Agnis, Dalmevin, Equa, Galvus, Glucemix, Proglin, Viglita, Viptin, Xiliarx, Zavamet, Zomelis) Alogliptin (Nesina) Saxagliptin (Onglyza) Meglitinides Nateglinide (Glinate, Glytan, Incuria, Inglex, Starlix) Repaglinide (Enyglid, Eurepa, Prandin, Novonorm, Surepost) Insulin injections Insulin Glulisine (Apidra) Insulin Lispro (Humalog) Insulin Aspart (Novolog, Novomix, NovoRapid) Glimepiride + Pioglitazone (Piozer G) Dapagliflozin (Edistride, Glycema) Insulin regular (Humulin R, Mixredlin, Novolin ge Toronto, Hypurin regular/neutral, Actrapid) Metformin + Glimepiride + Pioglitazone (Tripride) Metformin + Rosiglitazone (Avandamet) Metformin + Vidagliptin (Galvus Met) Insulin Isophane, NPH (Humulin N, Hypurin NPH, Isophan, Insuman, Insulatard, Novolin ge NPH, Novolin N, Protaphane) Insulin Lente/Insulin Zinc (Humulin L) Insulin Glargine/Glargine U-300 (Basaglar, Lantus, Lusduna, Optisulin, Toujeo, Toujeo SoloSTAR)

Supplementary Table 1. Continued

Medication group	Generic names
	Insulin Detemir (Levemir) Insulin Ultralente/Extended Insulin Zinc (Humulin U)
Nitrates	Glyceryl Trinitrate/Trinitroglycerin/Nitroglycerine/TNG (Anginil, Angised, Anril sr, Cardiotec, Corcont, Minitran, Nitrostat) Isosorbide dinitrate (Cardiket, Coronex, Dilatrate SR, Isordil, Isordil Titradose, Kardicket, Onset, Sorbitrate) Isosorbide mononitrate (Imdur, ISMN, Monoket) Nitroglycerin spray (Nitro spray, Nitromist) Propatyl nitrate (Dainitre, Sustrate)
α antagonists	Prazosin (Minipress, Apo-prazo, Nu-prazo) Alfuzosin (Alfusin, Apo-Alfuzosin, Sandoz-Alfuzosin, Uroxatral, Xatral) Terazosin (Apo-terazosin, Blavin, Dom-terazosin, Hytrin, Hytrinex, Prostatil, Teranin, Teva-terazosin) Tamsulosin (Apo-Tamsulosin, Diffundox, Dynapres, Flomax, Maxflow, Modalusin, Omnic, Omnic XL, Ratio-Tamsulosin, Reduprost, Sandoz-Tamsulosin, Secotex, Tamprost, Tamprost-MR, Urilosin, Veltam) Doxazosin/Doxazosin Mesylate (Cadex, Cascor, Cardura, Doxadura, Europrostatin, Mesidox) Silodosin (Rapaflo, Urorec)
Angiotensin II receptor blockers	Olmesartan (Benicar, Olmecip, Olmetec, Olmetor) Telmisartan (Micardis, Telday, Telista) Candesartan (Amias, Atacand, Blopress, Candesar, Ratacand) Irbesartan (Avapro, Xarb, Irovel) Losartan (Cozaar, Eziday, Losar, Nusar, Tozaar) Valsartan (Diovan, Prexxartan, Valent, Valsan, Valzaar, Valz) Azilsartan (Edarbi, Macarbi)
Antituberculosis medications	Ethambutol + Isoniazid + Pyrazinamide + Rifampicin (AKT4, Rimstar, Voractiv) Rifampicin + Isoniazid + Pyrazinamide (Binex-Z, Rifater, Tricox) Capreomycin (Capastat, Kapocin, Lykocin) Rifabutin/Ansamycin (Ansatipin, Macbutin, Mycobutin, Ributin, Tambux) Cycloserine (Seromycin) Ethambutol/EMB (Myambutol) Isoniazid (LUP-INH, Nicozid, Nydrazid, Tibinide) Pyrazinamide (Piraldina, Prazina, Pyrazide, Tebrazid) Terizidone (Tericox, Terivalidin)
Antiarrhythmic medications	Quinidine/Conquinine/Pitayine (Cardioquin, Kinidin, Natcardine, Quinate, Wanidine) Procainamide (Biocoryl, Procan SR, Pronestyl) Amiodarone (Aratac, Arycor, Cordarone, Pacerone, Ritmocardyl) Adenosine (Adenocard, Adenoject, Adenoscan) Disopyramide (Rythmodan, Norpace) Propafenone (Rythmol, Rytmonorm) Flecainide (Tambocor) Dronedarone (Multaq)
Blood thinners	Pentoxifylline/Oxpentifylline (Angiopent, Elorgan, Rentylin, Trental, Pentoxil) Prasugrel (Effient, Efient, Prasugrin) Ticlopidine (Ticlogard, Ticlid, Ticlop) Ticagrelor (Brilinta) Heparin Dipyridamole (Persantine, Tromboliz) Enoxaparin (Clexane, Lovenox, Xaparin) Dabigatran (Pradax, Pradaxa) Edoxaban (Savaysa) Apixaban (Eliquis) Rivaroxaban (Xarelto) Anagrelide (Agrylin)
NSAIDs	Ibuprofen (Advil, Brufen, Ibugesic, Motrin, Nurofen) Ketoprofen (Apo Ketoprofen, Actron, Arveles, Bi-Profenid, Orudis, Profenid) Fenoprofen (Nalfon, Fenortho) Piroxicam (Feldene, Pirox, Pirosil, Oksikam) Meloxicam (Acticam, Anjeso, Articam, Exen Fort, Flexol, Meksun, Melflam, Melkam, Melic, Melox, Muvera, Mobic, Romacox fort, Runomex Fort) Tenoxicam (Mobiflex, Okasmen, Tobitil) Naproxen (Aleve, Anaprox, Apranax, Aprol, Artagen, Flanax, Midol extended relief, Naprelan 500, Naprosyn) Indomethacin/Indometacin (Apo-Indomethacin, Endomed, Indocid, Indocin, Indomethol, Tivorbex)

Supplementary Table 1. Continued

Medication group	Generic names
	Oxaprozin (Apo Oxaprozin, Daypro, Duraprox) Mefenamic Acid (Fenamin, Mefac, Ponstel, Prostan) Sulindac (Clinoril) Valdecoxib (Bextra, Creval) Etoricoxib (Arcoxia, Nucoxia, Etozox, Etrobax) Celecoxib (Articoxib, Celebrex, Cobix, Celact, Celetop) Aceclofenac (Acepron, Aclofen, Biofenac) Diclofenac (Cataflam, Diclo-Denk, Dicloflam, Diclomec, Dikloron, Catafen, Kalidren, Rofenac, Romatin, Voveran, Voltaren, Zipsor, Zorvolex) Ketorolac (Acular, Pevoran, Toradol) Diflunisal (Dolobid) Tolfenamic acid (Clotam, Flocur rapid) Phenylbutazone (Azolid, Butazolidin) Nimesulide (Acinimide-MD, Adenim, Aglonim, Algifeno, Arslide, Artilide, Bangni, Cimelide, Dolvex, Mesulid, Nimesil, Nimex, Nise) Etodolac (Etol Fort, Lodine, Tadolak) Etofenamate (Flexo Ampul, Thermo-Efemat) Flurbiprofen (Ocufen, Majezik) Lornoxicam (Camri, Xefo Rapid) Acetamin (Acetudil, Rantudil, Shun Song, Tendonil, Zadex) Tiaprofenic acid (Surgam, Surgamyl, Tiaprofen)
Antibiotics	Penicillin Penicillin G/Benzylpenicillin (Crystapen, Pfizerpen, Penidure) Penicillin V/Phenoxymethylpenicillin (Apocillin, Beepen-VK, Veetids) Amoxycillin (Amoval, Amoxil, Amoxicot, Aristomox, Atoksilin, Blumox, Moxatag, Moxilin, Trimox) Ampicillin (Alfasilin, Ampi-Co, Principen, Roscillin, Zycilin) Flucloxacillin (Heracillin) Amoxycillin + clavulanic acid/Clavulanate (Amocladan, Apo-Amoxi Clav, Augmentin, Calamox, Clavam, Croxilex, Croxilex-Bid, Klamoks, Klavupen) Cephalosporins Cefadroxil (Cefadrox, Duricef, Droxyl, Neucef) Cephalexin (Keflex, Phexin, Sporidex) Cefazolin (Cezolin, Reflin) Cefaclor (CEC, Ceclor, Distaclor, Keflor) Cefuroxime (Akasef, Ceftin, Enfexia, Xylid, Zocef) Cefdinir (Aldinir, Omnicef, Zinir) Cefixime (Ceftas, Fixx, Suprax) Cefotaxime (Taxim, Taxobid) Cefpodoxime (Cefoprox, Cepodem, Vantin) Ceftriaxone (Oframax, Rocephin, Torocef) Cefoperazone (Cefomycin, Cefobid, Magnamycin) Ceftazidime (Fortaz, Fortum, Orzid, Tazicef, Tizime) Cefepime (Maxipime, Megapime) Cefpirome (Cefir) Loracarbef (Lorabid, Lorafem, Lorbef) Cefradine (Cefalmin) Cefprozil (Brisoral, Cefzil, Serozil) Cefditoren (Meiact, Spectracef) Aminoglycosides Gentamycin (Garamycin, Genticyn) Tobramycin (Tobacin, Tobraneg) Neomycin/Fradiomycin (Neo-Fradin, Mycifradin) Netilmycin/Netilmicin (Netira, Netromicina, Netromycin) Amikacin (Amikin) Streptomycin (Ambistryn-S, Cipstryn) Kanamycin (Kantrex) Tetracyclines Doxycycline (Doxin, Doxy-Caps, Periostat) Tetracycline (Cadicycline) Oxytetracycline (Terramycin) Lymecycline/Lymecyclinum/tetracycline-L-methylene lysine (Eficiclina/Tetralisal/Tetralysal) Minocycline (Dynacin, Minocin, Minolira, Vectrin)

Supplementary Table 1. Continued

Medication group	Generic names
	Macrolides Erythromycin (Althrocin, Erythrocin) Azithromycin (Azee, Azeltin, Azicare, Azipro, Azithral, Macrotor, Zimak, Zythromax) Clarithromycin (Clarimac, Clamycin, Cleanomicin, Klacid, Klamaxin, Klaritromicin, Laricid, Maclar, MacRol) Roxithromycin (Roxil, Roximol) Telithromycin (Ketek) Quinolones Ciprofloxacin (Baycip, Cetraxal, Cifran, Ciloxan,Ciphin, Ciplox, Ciprinol, Cipro, Ciprobid, Ciprolex, Cipronatin, Cipronex, Ciproxin, Ciprozid, Floxacin, Floxantina, Floxamin, Novidat) Ofloxacin (Aroflox, Curitol, Gyrex, Oflin, Tarivid, Zanocin) Levofloxacin (Loxof, Selan Tavanic, Alevo) Sparfloxacin (Bluspar, Flospar) Gemifloxacin (Factive, G Floren, Gemif, Gemitab, Topgem) Moxifloxacin (Moxicip, Avalox, Avelox) Norfloxacin/Norfloxacine (Lexinor, Noroxin) Carbapenem Ertapenem (INVanz) Meropenem (Meronem, Zaxter) Lincomycin Clindamycin (Cleocin, Cleocin Pediatric) Fosfomycin (Ivozfo, Monurol, Veramina) Other antibiotic medications Linezolid (A-Lide, Anozilad, Arlin, Linez, Zizolid, Zodlin, Zyvox, Zyvoxam) Ethionamide (Trecator) Rifampicin/Rifampin (Cavidin, Rifadin, Rofact)
Oral contraceptives	Estradiol/Ethinyl Estradiol (Climara, Estrace, Estrogel, Estraderm, Estradot, Estring, Estreva, Kilogest, Natifa, Oesclim, Oestrogel, Progyluton, Progynova, Primogyna, Trigestam, Vagifem, Vivelle Dot) Estriol (Orgestriol, Colpoestriol) Estrone (Estrona) Estropipate (Harmogen, Ortho-Est) Etonogestrel (Implanon) Conjugated estrogen - CES (Premarin) Medroxyprogesterone (Medroxy, Farlutal, Novo Medrone, Apo-Medroxy, Alti-Mpa, Provera) Desogestrel (Cerazette, Azalia) Progesterone (Prometrium) Norgestrel (Ovrette) Levonorgestrel (Microlut, Mirena) Tibolone (Climatix, Libiam, Livial, Sibolone, Tibofem, Tinox) Lynestrenol (Exluton, Orgametril, Ministat) Norethisterone/Norethindrone (Micronor, Primolut, Camila)
Proton pump inhibitors	Omeprazole (Acimed, Encid, Gasec, Gastec, Losec, Mylan-Omeprazole, Omzed, Novo-Omeprazole, Omecap, Omee, Omel, Omecip, Omenix, Omepracid, Omeprol, Omep, Ometid, Omex, Omez, Omezol, Omitin, Opal, Procelac, Proceptin, Ratio-Omeprazole, Sandoz-Omeprazole, Risek, Ruling, Seclo, Ulciprom, Zenpro) Pantoprazole (Mylan-Pantoprazole, Novo-Pantoprazole, Pantoloc, Pantaloc, Pan 40, Pantop, Panet, Pangest, Pantobex, Pantocid, Pantozol, Pantus, Pantosec, Protonix, Ratio-Pantoprazole, Sandoz-Pantoprazole, Tecta, Teva-Pantoprazole, Zopan) Lansoprazole (Aprasol, Prevacid, Degastrol, Lanasis, Lansazol, Lanzopral, Lansor, Lanzedin, Novo-Lansoprazole, Opagis, Selanz-SR, Teva-Lansoprazole, Zoprol) Rabeprazole (Pariet, Novo-Rabeprazole, Rabec, Rabekind, Rabelin, Rabelis, Rabezol, Rabicip, Rabicon, Rabilife, Rabistil, Rabium, Rablet, Rabonik, Ropraz, Rasonix, Razo, Repral, Roppi, Sandoz-Rabeprazole, Teva-Rabeprazole, Veloz) Esomeprazole (Esoral, Espra, Maxpro, Mylan-esomeprazole, Nexium, Sergel, Sompraz) Dexlansoprazole (Dexilant)
Antimalarial medications	Chloroquine/Chloroquine phosphate (Novo-Chloroquine, Resochin) Quinine/Quinine Sulphate (Apo Quinine, Qualaquin, Teva Quinine, Novo Quinine, Pro Quinine) Hydroxychloroquine/Hydroxychloroquine Sulphate (Hydroxyquine, Apo Hydroxyquine, Plaquenil, Reuquinol) Atovaquone (Mepron) Artesunate (Falcigo, Medisonate) Artemisinin/Dihydroartemisinin/Artenimol (Alaxin) Halofantrine (Halfan) Pyrimethamine (Daraprim)

Supplementary Table 1. Continued

Medication group	Generic names
	Mefloquine (Lariam) Primaquine (Neo-Quipenyl, Primachin) Proguanil (Paludrine)
Anticonvulsant medications	Carbamazepine (Carbatrol, Epitol, Hermolepsin, Karbalex, Karberol, Ketex, Tegretol, Teril, Zeptol) Valproate sodium/valproic acid/Divalproex sodium (Absenor, Convulex, Dapakan, Dayo OD, Dicorate, Diproex, Encorate, Epival, Depakine, Depakote, Torval, Valparin) Phenytoin (Dilantin, Ditoin, Epamin, Epanutin, Epdantoin, Eptoin, Fenigramon, Fenitoina, Hidantal) Phenobarbital/phenobarbitone (Gardenal, Luminal) Lamotrigine (Epiral, Flamus, Lamictal, Lamitor, Lamirax) Levetiracetam (Levetrol, Keppra, Levipil, Molival, Roweepra, Roweepra XR, Spritam) Lacosamide (Vimpat) Methsuximide/Mesuximide (Celontin, Petinutin) Ethosuximide (Zarontin) Oxcarbazepine (Oxetol, Trileptal) Primidone (Mysoline) Gabapentin (Fanatrex, Gabarone, Gralise) Topiramate (Tamate, Topamax, Topictal, Trokendil) Zonisamide (Excegran, Zonegran) Pregabalin (Lyrica, Nervax, Neugaba, Teural) Felbamate (Allace, Felbamyl, Felbatol, Taloxa) Barbexaclone/chp-Phenobarbitalat (IS) (Maliasin)
Antihistamine medications	Desloratadine (Aerius, Clarinex, Deswin) Loratadine (Aerotina, Allertin, Claritin, Claratyne, Fristamin, Lomilan, Nularef, Sinaler, Tislor) Cetirizine (Alatrol, Alerid, Aller-Relief, Amertil, Cetrizet, Finallerg, Letizen, Okacet, Reactine, Simtec, Zyrtec) Levocetirizine (Lerafil, Levocet, Levosiz, Levrix, Sinalerg, Vozet, Xyzal) Fexofenadine (Alercas, Allegra, Fexadine, Fexet, Fexigra, Fexohist, Telfast) Diphenhydramine (Allermax, Banophen, Benadryl, Diphenhist, Dorm-Aide, Dytuss, Gayaben, Nytol, UnisomZendryl, Vanamine PD, Z-Sleep, ZzzQuil) Hydroxyzine/Hidroxicina (Atarax, Hixizine, Hyzine, Rezine, Riva-Hydroxyzin) Pheniramine (Avil) Chlorpheniramine/Chlorphenamine (Ahiston, Chlor-Tripolon, Piriton) Dexchlorpheniramine (Polaramine) Chlorcyclizine (Zimin) Triprolidine (Actidil, Histex, Zymine) Rupatadine (Rupafin, Rupax, Urtimed) Olapatadine (Winolap) Promethazine (Histandil, Phenergan) Ebastine (Ebast, Kestine, Kestine Lyo) Ketotifen (Apo-ketotifen, Novo-ketotifen, Nu-ketotifen, PMS-ketotifen, Tofen, Zaditen) Clemastine/Meclastin (Tandegyl,Tavegil, Tavegyl, Tavist, Tavist Allergy) Alimemazine/Methylpromazine/Trimeprazine (Panectyl, Theralen, Vallergan) Doxylamine (Unisom) Carbinoxamine (Clistin, Karbinal, Mentex syrup) Azelastine (Astelin)
Hormonal medications	Estradiol/Ethinyl Estradiol (Climara, Estrace, Estrogel, Estraderm, Estradot, Estring, Estreva, Kilogest, Natifa, Oesclim, Oestrogel, Progyluton, Progynova, Primogyna, Trigestam, Vagifem, Vivelle Dot) Estriol (Orgestriol, Colpoestriol) Estrone (Estrona) Estropipate (Harmogen, Ortho-Est) Etonogestrel (Implanon) Conjugated Estrogen - CES (Premarin) Medroxyprogesterone (Medroxy, Farlutal, Novo Medrone, Apo-Medroxy, Alti-Mpa, Provera) Desogestrel (Cerazette, Azalia) Progesterone (Prometrium) Norgestrel (Ovrette) Levonorgestrel (Microlut, Mirena) Tibolone (Climatix, Libiam, Livial, Sibolone, Tibofem, Tinox) Lynestrenol (Exluton, Orgametril, Ministat) Norethisterone/Norethindrone (Micronor, Primolut, Camila) Testosterone/Testosterone Undecanoate (pms-Testosterone, Androgel, Andriol, Androderm, Delatestryl, Testim)

Supplementary Table 1. Continued

Medication group	Generic names
	Prasterone/Dehydroepiandrosterone (DHEA, Intrarosa) Bromocriptine (Bromergon, Cycloset, Grifocriptina, Gynodel, Lactodel, Parlodel, Proctinal, Sicriptin) Flutamide (Grisetin) Cabergoline (Cabaser, Dostinex) Cyproterone Acetate (Androcur, Cyprostat)
Antiviral medications	Single-tablet regimens Efavirenz + Tenofovir + Emtricitabine (Atripla, Tribuss, Odimune) Rilpivirine + Tenofovir Disoproxil Fumarate + Emtricitabine (Complera) Elvitegravir + Tenofovir Alafenamide + Emtricitabine + Cobicistat (Genvoya) Rilpivirine + Emtricitabine + Tenofovir Alafenamide (Odefsey) Elvitegravir + Tenofovir Disoproxil Fumarate + Emtricitabine + Cobicistat (Stribild) Dolutegravir + Abacavir + Lamivudine (Triumeq) Nucleoside/nucleotide reverse transcriptase inhibitors Emtricitabine (Emtriva) Lamivudine (Epivir, Heptovir, Zeffix) Zidovudine/Azidothymidine (Retrovir) Didanosine (Videx EC) Tenofovir Disoproxil Fumarate (Viread) Tenofovir Alafenamide (Vemlidy) Stavudine/Sanilvudine (Lion, Zerit) Abacavir (Ziagen) Zidovudine + Lamivudine (Combivir) Abacavir + Lamivudine (Epzicom, Kivexa) Abacavir + Zidovudine + Lamivudine (Trizivir) Tenofovir Disoproxil Fumarate + Emtricitabine (Truvada) Adefovir dipivoxil (Adenofovir, Hepsera) Ribavirin/Tribavirin (Copegus, Ibavyr, Rebetol, RibaPak, Riva tab, Vilona, Virazide, Virazole, Viron) Non-nucleoside reverse transcriptase inhibitors Rilpivirine (Edurant) Etravirine (Intelence) Delavirdine (Rescriptor) Efavirenz (Stocrin, Sustiva) Nevirapine (Viramune, Viramune XR) Protease inhibitors Tipranavir (Aptivus) Indinavir (Crixivan) Atazanavir (Reyataz) Ritonavir (Empetus, Norvir) Atazanavir + Cobicistat (Evotaz) Lopinavir + Ritonavir (Kaletra, Aluvia) Other antiretroviral medications Maraviroc (Celsentri) Enfuvirtide (Fuzeon) Anti-influenza medications Amantadine (Amantrel, Gocovri, Osmolex ER, Symmetrel) Other antiviral medications Acyclovir (Zovirax) Inosine/Isoprinosine/Methisoprinol (Imunovir) Valacyclovir (Apo-valacyclovir, Co-valacyclovir, Valtrex)

Supplementary Table 2. Medication Use Among the Study Population Among Crohn's Disease Patients

	Short-term (baseline use)		Long-term (baseline and follow-up evaluation)	
Medication, n (%)	Overall (n = 133,137)	Participants who developed CD (n = 143)	Overall (n = 133,137)	Participants who developed CD (n = 143)
Angiotensin-converting enzyme inhibitors	5861 (4.4)	7 (4.9)	2656 (2.0)	4 (2.8)
β-blockers	5027 (3.8)	8 (5.6)	2269 (1.7)	2 (1.4)
Calcium antagonists	4660 (3.5)	2 (1.4)	1549 (1.2)	1 (0.7)
Diuretics	5272 (4.0)	5 (3.5)	1531 (1.1)	2 (1.4)
Statins	4611 (3.5)	17 (11.9)	2324 (1.7)	8 (5.6)
Aspirin	5323 (4.0)	13 (9.1)	1985 (1.5)	5 (3.5)
Clopidogrel	301 (0.2)	0	43 (0.1)	0
Vitamin K antagonists	587 (0.4)	0	123 (0.1)	0
Oral/injectable hypoglycemic medications	5232 (3.9)	7 (4.9)	2343 (1.8)	3 (2.1)
Nitrates	660 (0.5)	0	121 (0.1)	0
α Antagonists	367 (0.3)	1 (0.7)	19 (0.1)	2 (1.4)
Angiotensin II receptor blockers	2424 (1.8)	7 (4.9)	968 (0.7)	0
Antituberculosis medications	63 (0.1)	0	1 (0.1)	0
Antiarrhythmic medications	189 (0.1)	3 (2.1)	48 (0.1)	0
Blood thinners	747 (0.6)	0	80 (0.1)	0
NSAIDs	2939 (2.2)	5 (3.5)	147 (0.1)	1 (0.7)
Antibiotics	947 (0.7)	2 (1.4)	25 (0.1)	0
Oral contraceptives*	24,610/79,142 (31.1)	36 (25.2)		
Proton pump inhibitors	294 (0.2)	1 (0.7)		
Antimalarial medications	68 (0.1)	2 (1.4)	1 (0.1)	0
Anticonvulsant medications	99 (0.1)	0	1 (0.1)	0
Antihistamine medications	104 (0.1)	0	0	0
Hormonal medications	405 (0.3)	1 (0.7)	0	0
Antiviral medications	47 (0.1)	0		

NSAID, nonsteroidal anti-inflammatory drug.
*Among females.

Supplementary Table 3. Medication Use Among the Study Population Among Ulcerative Colitis Patients

	Short-term (baseline use)		Long-term (baseline and follow-up evaluation)	
Medication, n (%)	Overall (n = 133,137)	Participants who developed UC (n = 428)	Overall (n = 133,137)	Participants who developed UC (n = 428)
Angiotensin-converting enzyme inhibitors	5861 (4.4)	26 (6.1)	2656 (2.0)	17 (4.0)
β-blockers	5027 (3.8)	29 (6.8)	2269 (1.7)	12 (2.8)
Calcium antagonists	4660 (3.5)	7 (1.6)	1549 (1.2)	4 (0.9)
Diuretics	5272 (4.0)	23 (5.4)	1531 (1.1)	12 (2.8)
Statins	4611 (3.5)	30 (7.0)	2324 (1.7)	17 (4.0)
Aspirin	5323 (4.0)	33 (7.7)	1985 (1.5)	13 (3.0)
Clopidogrel	301 (0.2)	2 (0.5)	43 (0.1)	0
Vitamin K antagonists	587 (0.4)	2 (0.5)	123 (0.1)	2 (0.5)
Oral/injectable hypoglycemic medications	5232 (3.9)	26 (6.1)	2343 (1.8)	11 (2.6)
Nitrates	660 (0.5)	7 (1.6)	121 (0.1)	1 (0.2)
α Antagonists	367 (0.3)	1 (0.2)	19 (0.1)	0
Angiotensin II receptor blockers	2424 (1.8)	17 (4.0)	968 (0.7)	8 (1.9)
Antituberculosis medications	63 (0.1)	1 (0.2)	1 (0.1)	0
Antiarrhythmic medications	189 (0.1)	2 (0.5)	48 (0.1)	0
Blood thinners	747 (0.6)	2 (0.5)	80 (0.1)	0
NSAIDs	2939 (2.2)	24 (5.6)	147 (0.1)	4 (0.9)
Antibiotics	947 (0.7)	14 (3.3)	25 (0.1)	0
Oral contraceptives*	24,610/79,142 (31.1)	149 (34.8)		
Proton pump inhibitors	294 (0.2)	0		
Antimalarial medications	68 (0.1)	0	1 (0.1)	0
Anticonvulsant medications	99 (0.1)	1 (0.2)	1 (0.1)	0
Antihistamine medications	104 (0.1)	1 (0.2)	0	0
Hormonal medications	405 (0.3)	4 (0.9)	0	0
Antiviral medications	47 (0.1)	1 (0.2)		

NSAID, nonsteroidal anti-inflammatory drug.
*Among females.

Supplementary Table 4. Association Between Baseline Medication Use and Development of Crohn's Disease

	Unadjusted		Minimally adjusted		Fully adjusted	
Medication	OR (95% CI)	*P* value	OR (95% CI)	*P* value	OR (95% CI)	*P* value
Angiotensin-converting enzyme inhibitors	1.12 (0.52–2.39)	.771341	0.77 (0.36–1.67)	.51309	0.89 (0.36–2.24)	.811961
β-blockers	1.52 (0.74–3.09)	.253938	0.96 (0.47–1.97)	.903537	0.98 (0.39–2.45)	.963504
Calcium antagonists	0.39 (0.10–1.58)	.186301	0.33 (0.08–1.33)	.119037	0.25 (0.03–1.81)	.169677
Diuretics	0.88 (0.36–2.15)	.77838	0.80 (0.33–1.97)	.628977	1.06 (0.38–2.99)	.90714
Statins	3.78 (2.28–6.28)	2.72E-07	2.35 (1.39–3.96)	.001409	2.15 (1.11–4.18)	.023542
Aspirin	2.41 (1.36–4.27)	.002499	1.74 (0.97–3.13)	.062446	1.91 (0.96–3.79)	.063406
Clopidogrel	N/A	N/A	N/A	N/A	N/A	N/A
Vitamin K antagonists	N/A	N/A	N/A	N/A	N/A	N/A
Oral/injectable hypoglycemic medications	1.26 (0.59–2.70)	.5506	0.93 (0.43–2.00)	.8475	0.84 (0.30–2.39)	.7498
Nitrates	N/A	N/A	N/A	N/A	N/A	N/A
α Antagonists	2.55 (0.36–18.28)	.351438	1.40 (0.19–10.16)	.741023	1.79 (0.24–13.24)	.567115
Angiotensin II receptor blockers	2.79 (1.30–5.97)	.008167	1.42 (0.66–3.08)	.369772	1.23 (0.44–3.41)	.692731
Antituberculosis medications	N/A	N/A	N/A	N/A	N/A	N/A
Antiarrhythmic medications	15.42 (4.87–48.82)	3.3E-06	10.47 (3.27–33.50)	7.61E-05	5.85 (0.80–42.87)	.082395
Blood thinners	N/A	N/A	N/A	N/A	N/A	N/A
NSAIDs	2.21 (0.90–5.42)	.08468	1.49 (0.61–3.69)	.382789	1.49 (0.60–3.70)	.385976
Antibiotics	2.72 (0.67–11.03)	.161802	1.80 (0.44–7.33)	.410717	1.71 (0.42–6.97)	.433569
Oral contraceptives*	2.92 (1.72–4.96)	7.67E-05	2.39 (1.30–4.09)	.001435143	1.95 (1.11–3.42)	.020624
Proton pump inhibitors	3.18 (0.44–22.80)	.249886	7.81 (1.06–57.30)	.043238	10.71 (1.41–81.44)	.02192
Antimalarial medications	28.48 (6.91–117.39)	3.58E-06	86.47 (20.13–371.39)	2E-09	84.74 (11.09–647.52)	1.88E-05
Anticonvulsant medications	N/A	N/A	N/A	N/A	N/A	N/A
Antihistamine medications	N/A	N/A	N/A	N/A	N/A	N/A
Hormonal medications	2.33 (0.32–16.67)	.400617	4.96 (0.68–36.28)	.114781	6.17 (0.82–46.45)	.074913
Antiviral medications	N/A	N/A	N/A	N/A	N/A	N/A

N/A, not applicable; NSAID, nonsteroidal anti-inflammatory drug; OR, odds ratio.
*Among females.

Supplementary Table 5. Association Between Baseline Medication Use and Development of Ulcerative Colitis

	Unadjusted		Minimally adjusted		Fully adjusted	
Medication	OR (95% CI)	*P* value	OR (95% CI)	*P* value	OR (95% CI)	*P* value
Angiotensin-converting enzyme inhibitors	1.41 (0.95–2.09)	.092587	1.10 (0.73–1.65)	.644513	1.01 (0.65–1.57)	.979485
β-blockers	1.86 (1.27–2.71)	.001315	1.37 (0.93–2.00)	.111776	1.38 (0.92–2.06)	.118426
Calcium antagonists	0.46 (0.22–0.97)	.040212	0.43 (0.20–0.90)	.026311	0.37 (0.16–0.85)	.018189
Diuretics	1.38 (0.91–2.10)	.134802	1.35 (0.88–2.07)	.166495	1.37 (0.86–2.16)	.183048
Statins	2.11 (1.46–3.07)	8.1E-05	1.48 (1.01–2.17)	.042678	1.27 (0.84–1.92)	.261054
Aspirin	2.02 (1.41–2.88)	.000115	1.63 (1.13–2.34)	.009131	1.59 (1.09–2.31)	.016384
Clopidogrel	2.08 (0.52–8.37)	.303982	1.27 (0.31–5.11)	.741533	0.58 (0.08–4.19)	.59302
Vitamin K antagonists	1.06 (0.26–4.26)	.935439	0.56 (0.14–2.26)	.417324	0.52 (0.13–2.11)	.362193
Oral/injectable hypoglycemic medications	1.45 (0.96–2.20)	.0751	1.21 (0.79–1.83)	.3793	1.00 (0.62–1.62)	.9985
Nitrates	3.36 (1.59–7.12)	.001564	3.29 (1.54–7.02)	.002028	3.58 (1.65–7.77)	.001248
α antagonists	0.85 (0.12–6.05)	.869599	0.54 (0.08–3.89)	.542013	0.54 (0.08–3.88)	.540953
Angiotensin II receptor blockers	2.24 (1.38–3.65)	.001141	1.43 (0.87–2.34)	.156529	1.24 (0.73–2.10)	.433059
Antituberculosis medications	5.09 (0.70–36.77)	.106934	7.64 (1.05–55.65)	.044689	7.59 (0.97–59.19)	.053099
Antiarrhythmic medications	3.38 (0.84–13.65)	.087608	2.58 (0.64–10.46)	.184646	2.90 (0.71–11.84)	.136989
Blood thinners	0.83 (0.21–3.34)	.793845	1.31 (0.32–5.28)	.70603	0.76 (0.11–5.50)	.78797
NSAIDs	2.69 (1.78–4.07)	2.69E-06	1.90 (1.25–2.88)	.002581	1.89 (1.24–2.87)	.002936
Antibiotics	4.86 (2.84–8.32)	7.67E-09	3.45 (2.01–5.92)	6.54E-06	3.13 (1.79–5.48)	6.42E-05
Oral contraceptives*	2.80 (2.16–3.62)	5.40E-15	2.41 (1.85–3.13)	4.42E-11	2.22 (1.70–2.91)	9.32E-09
Proton pump inhibitors	N/A	N/A	N/A	N/A	N/A	N/A
Antimalarial medications	N/A	N/A	N/A	N/A	N/A	N/A
Anticonvulsant medications	3.17 (0.44–22.75)	.252132	5.28 (0.73–38.04)	.09898	5.57 (0.76–41.06)	.09181
Antihistamine medications	3.01 (0.42–21.63)	.273087	5.41 (0.75–39.07)	.093884	4.82 (0.64–36.10)	.125967
Hormonal medications	3.12 (1.16–8.38)	.02432	4.86 (1.79–13.18)	.001891	4.19 (1.52–11.58)	.005385
Antiviral medications	6.75 (0.93–49.03)	.05923	11.85 (1.62–86.55)	.014829	13.60 (1.80–102.91)	.011468

N/A, not applicable; NSAID, nonsteroidal anti-inflammatory drug; OR, odds ratio.
*Among females.

Supplementary Table 6. IBD Prevalence in 1990 and 2017 for Both Sexes and Percentage Change in Age-Standardized Rates

Country	1990 counts (95% UI	2017 counts (95% UI)	Percentage change in age-standardized rates between 1990 and 2017
Global	3,699,934 (3,536,764–3,888,255)	6,848,872 (6,421,390–7,304,419)	6.1 (3.3–8.6)
Argentina	11,959 (10,847–13,249)	21,163 (19,583–23,187)	19.3 (7.9–31.3)
Bangladesh	8164 (7401–9027)	23,839 (21,203–27,231)	52.6 (35–71.3)
Brazil	72,271 (65,303–78,974)	107,647 (98,592–117,714)	-28.1 (-33.9 to -21.6)
Canada	17,538 (16,014–19,554)	20,360 (18,722–22,566)	-18.4 (-26.9 to -9.1)
Chile	5282 (4779–5850)	10,584 (9629–11,770)	14.8 (4.1–27.8)
China	1,047,992 (981,582–1,116,181)	2,665,081 (2,433,419–2,900,745)	34.5 (29.7–39.6)
Colombia	4873 (4457–5507)	14,058 (12,650–16,464)	19.5 (6.6–37.9)
India	89,149 (82,262–96,873)	212,451 (192,979–235,104)	28.6 (23.8–33.5)
Iran	9677 (8774–10,688)	23,812 (21,600–26,364)	16.2 (6.0–26.7)
Kazakhstan	5845 (5385–6319)	5777 (5332–6260)	-17.5 (-23.2 to -11.8)
Kyrgyzstan	1093 (1010–1184)	1792 (1664–1925)	5.3 (-1.9 to 12.6)
Malaysia	2069 (1896–2282)	5324 (4807–6110)	22.9 (11.2–39.6)
Pakistan	10,766 (9802–11,837)	26,936 (24,106–29,871)	14.3 (6.9–21.6)
Palestine	446 (351–574)	1161 (933–1445)	-13.2 (-23.5 to -0.3)
Philippines	5115 (4722–5589)	12,111 (11,183–13,067)	20.9 (12.4–29.4)
Poland	72,792 (67,488–78,845)	97,335 (90,624–104,919)	11.6 (6.0–17.8)
Russia	258,284 (232,383–284,795)	197,640 (179,945–216,250)	-32.1 (-38.7 to -25.5)
Saudi Arabia	2899 (2601–3294)	9375 (8370–11,115)	6.3 (-5.6 to 21.7)
South Africa	5392 (4922–5909)	5954 (5404–6519)	-39.4 (-43.0 to -35.5)
Sweden	10,413 (9155–12,393)	12,231 (10,520–15,016)	-0.7 (-16.4 to 20.5)
Tanzania	1406 (1314–1500)	3811 (3558–4090)	26.8 (20.3–33.0)
Turkey	16,031 (14,499–18,032)	38,881 (35,173–43,130)	30.1 (15.8–44.6)
United Arab Emirates	511 (386–669)	3682 (2607–5186)	-7.5 (-19.6 to 11.5)
Zimbabwe	925 (857–995)	966 (901–1033)	-33.1 (-36.5 to -29.5)

UI, uncertainty interval.
Adapted from Alatab et al.[8]